

Architectural Diversity in Multi-Agent LLM Deliberation

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Abstract

Large language models frequently produce confident but incorrect answers, especially on reasoning or fact-intensive tasks. Previous work (Du et al., 2023) has shown that multi-agent debate can improve factuality and reasoning of models by having multiple model instances independently answer a question, review and critiques each other’s answers, and converge on a final response. In this project, we study whether this debate workflow is more effective when the agents are architecturally diverse rather than identical. Specifically, we compare single-agent baselines, homogeneous debate between two instances of the same model, and heterogeneous debate between GPT-4o-mini and Gemini 2.5 Flash on difficult MATH-500 Level 4 and Level 5 problems. Our results suggest that homogeneous debate yields limited gains, while heterogeneous debate substantially improves the weaker model’s post-debate accuracy, though this effect may partly reflect exposure to a stronger model rather than diversity alone.

1 Motivation

As Large language models (LLMs) grow increasingly capable at natural language generation and complex reasoning, they still frequently make confident errors. These errors are concerning as we move towards using LLMs to chain reasoning, where ultimately each reason the model makes in the chain may contain subtle mistakes or factual hallucinations. Du et al. (Du et al., 2023) addressed this issue by proposing a multi-agent debate architecture: multiple LLM agents first generate independent answers, then view one another’s responses, revise their reasoning, and eventually converge on a final answer. Their core finding is that this process improves performance across reasoning and factuality benchmarks, including arithmetic, GSM8K, MMLU, biography factuality, and chess-related tasks

However, the original paper focused solely on a multi-agent setup consisting of multiple instances of the same model. This raises an important point of failure in the debate setup, as identical agents tend to make similar mistakes, then the debate provides limited value on mitigating such situations. Our project explores whether *model diversity* can improve multi-agent deliberation. Rather than using two copies of the same model, we compare homogeneous debate with heterogeneous debate between models from different families.

Our hypothesis is that heterogeneous debate can improve reasoning beyond homogeneous debate because different model families may fail differently. If one model identifies a reasoning path that another model misses, the debate process may allow the weaker or initially incorrect model to revise its answer. Alternatively, heterogeneous gains may also come from model capability differences, where if one model is much stronger at a specific domain, observed improvements may come from the weaker model simply adopting all of the answers of the stronger model, rather than using collaborative reasoning.

This paper makes three contributions. First, we implement a two-agent, two-round debate framework for level 4 and 5 MATH-500 problems. Second, we compare single-agent, homogeneous, and heterogeneous debate conditions across GPT-4o-mini and Gemini 2.5 Flash. Third, we analyze whether the observed gains support architectural diversity as a useful extension of multi-agent debate, while identifying limitations such as lack of true consensus, copying versus reasoning, and unequal model strength.

2 Related Work

Multi-agent debate. Du et al. (Du et al., 2023) propose using multiple LLM agents to independently answer a query, critique other agents’ an-

swers, and update their own response over several debate rounds, then converge to an agreed answer. Their work shows that debate improves both reasoning accuracy and factual validity. Our project builds directly on this framework, but focuses on a narrower question: whether agents from different model families produce better deliberation than agents from the same model family.

AI safety via debate. Irving et al. (Irving et al., 2018) introduced debate as a mechanism for making AI systems more interpretable and easier for humans to interpret. In their formulation, agents argue opposing sides and a judge determines which argument is more convincing. Our project differs because our agents are not adversarially assigned opposing positions. Instead, both agents attempt to solve the same math problem and use each other’s reasoning as additional evidence.

Self-consistency and majority voting. Self-consistency methods sample multiple reasoning chains from a model and select the most common answer (Wang et al., 2022). This is related to multi-agent debate because both approaches use multiple generated solutions. Critically, the self-consistency lacks the ability for agents to revise their reasoning after seeing alternatives, as each reasoning chain was finalized independently. Our project focuses on deliberative revision rather than only sampling and voting.

Chain-of-thought prompting. Chain-of-thought prompting improves reasoning by encouraging models to produce intermediate reasoning steps (Wei et al., 2022). Debate can be viewed as complementary to chain-of-thought. Instead of relying on one reasoning trace, multiple agents produce and revise each other’s traces. In our experiments, we use reasoning-oriented prompts so that agents expose intermediate reasoning for another model to assess and critique.

Project scope. Unlike Du et al., who evaluated their debate across six benchmarks and multiple agent/round configurations, we restrict our evaluation to difficult MATH-500 Level 4 and Level 5 problems in a 2-round debate due to time and compute constraints, as well as Du et al. observing limited reasoning gain after exceeding 2 rounds. This narrower scope lets us focus on the specific question of homogeneous versus heterogeneous debate while keeping the experiment reproducible.

3 Methods

3.1 Task and Dataset

We evaluate on MATH-500 Level 4 and Level 5 problems, which are challenging competition-style math problems. We chose this subset because hard reasoning problems are more likely to expose differences between single-agent, homogeneous debate, and heterogeneous debate. Each problem has a ground-truth final answer, allowing easy correctness evaluation by comparing the extracted final answer to the reference answer.

3.2 Models

We compare two model families:

- **OpenAI model:** GPT-4o-mini.
- **Google model:** Gemini 2.5 Flash.

These models were selected because they are accessible through API calls, relatively inexpensive compared to larger frontier models, and different enough to test whether cross-family model diversity helps deliberation.

3.3 Debate Architecture

Our debate framework uses two agents and two rounds. In Round 1, each agent receives the same math problem and independently produces a solution. In Round 2, each agent receives the other agent’s Round 1 response and is asked to revise or defend its answer. After Round 2, a final answer is selected.

For problem q , each agent a_i first generates:

$$r_i^{(1)} = M_i(q)$$

where M_i is the model assigned to agent i . Then each agent updates its answer using the other agent’s response:

$$r_i^{(2)} = M_i(q, r_j^{(1)})$$

where $j \neq i$. The final answer is extracted from the Round 2 output.

3.4 Experimental Conditions

We evaluate five conditions:

1. **Single OpenAI:** one GPT-4o-mini agent, no debate.
2. **Single Gemini:** one Gemini 2.5 Flash agent, no debate.

3. **Homogeneous OpenAI debate:** two GPT-4o-mini agents.
4. **Homogeneous Gemini debate:** two Gemini 2.5 Flash agents.
5. **Heterogeneous debate:** one GPT-4o-mini agent and one Gemini 2.5 Flash agent.

The single-agent settings provide no-debate baselines. The homogeneous conditions test whether debate helps when both agents come from the same model family and acts as a baseline for the next condition. The heterogeneous condition tests whether cross-family deliberation improves results.

3.5 Prompting Procedure

For each problem, the Round 1 prompt asks the model to solve the math problem step by step and place the final answer in a clearly extractable format, such as:

Solve the following math problem. Explain your reasoning clearly. Put your final answer at the end in the form `\boxed{answer}`.

For Round 2, the model is shown the other agent’s Round 1 response:

Another agent solved the same problem and gave the following response: [other agent response]. Review their reasoning carefully. If their solution reveals an error in your original answer, revise your answer. Otherwise, defend your answer. Put your final answer at the end in the form `\boxed{answer}`.

The same prompt template is used across model families to avoid giving one model an advantage through prompt differences.

3.6 Answer Extraction and Accuracy

We extract the final answer from each model response by searching for the final boxed answer. A response is marked correct if the extracted answer matches the dataset’s ground-truth answer after basic normalization, such as removing whitespace and equivalent formatting.

The primary metric is:

$$\text{Accuracy} = \frac{\text{Number of correctly answered problems}}{\text{Total number of evaluated problems}}$$

3.7 Additional Metrics

In addition to final accuracy, we track:

- **Disagreement rate:** how often the two agents end with different answers.

- **Agreement accuracy:** whether agents are more reliable when they agree.
- **Debate contribution:** whether debate fixes or breaks Agent 0’s original answer.
- **Cost and latency:** approximate API cost and runtime per sample for each setup.

These metrics help distinguish between genuine debate improvement and simple answer copying.

4 Results

4.1 Single-Agent and Homogeneous Debate Baselines

Table 1 summarizes the single-agent and homogeneous debate results.

Condition	Single-Agent Accuracy	Homogeneous Debate Accuracy
GPT-4o-mini	54%	58%
Gemini 2.5 Flash	86%	87%

Table 1: Single-agent and homogeneous debate results. Homogeneous debate produced only small improvements: +4 percentage points for GPT-4o-mini and +1 percentage point for Gemini 2.5 Flash.

The homogeneous results suggest that debate between identical model families provided limited gains on this hard subset. One explanation is that similar agents often fail on the same problems. When both agents start with the same incorrect reasoning pattern, the second round has little new information to use.

4.2 Heterogeneous Debate Results

Table 2 summarizes the heterogeneous debate results.

Agent in Heterogeneous Debate	Round 1 Accuracy	Round 2 Accuracy
Gemini 2.5 Flash	84%	85%
GPT-4o-mini	54%	84%

Table 2: Heterogeneous debate results between GPT-4o-mini and Gemini 2.5 Flash. GPT-4o-mini improved substantially after seeing Gemini’s reasoning, while Gemini changed very little.

The most striking result is that GPT-4o-mini improved from 54% in Round 1 to 84% in Round 2 when paired with Gemini 2.5 Flash. This gain was much larger than the gain observed in homogeneous GPT-4o-mini debate. The final heterogeneous debate accuracy reached approximately 85%, close to Gemini’s single-agent baseline and far above GPT-4o-mini’s single-agent baseline.

4.3 Interpretation

These results partially support our hypothesis that model diversity can improve debate. The heterogeneous pairing gave the weaker model access to reasoning that it did not produce independently, allowing it to correct many initial mistakes. However, the results also show an important issue: Gemini 2.5 Flash performed much better than GPT-4o-mini on this benchmark. Because of this, the improvement may not be only due to having more diverse answers. Instead, it may be because the weaker model followed or adopted the stronger model’s answer.

5 Discussion and Limitations

Our findings suggest that heterogeneous debate can be useful, especially when one model has reasoning that another model lacks. However, the experiment also exposes several limitations.

First, our setup uses only two agents. With two agents, disagreement creates a tie rather than a true consensus. This makes final answer selection more fragile and can bias results toward the designated tie-breaking agent.

Second, we cannot yet fully distinguish reasoning improvement from reasoning/answer copying. GPT-4o-mini’s large improvement after seeing Gemini’s response may mean that it learned from Gemini’s reasoning, but it may also mean that it copied Gemini’s final answer. A transcript-level analysis is needed to determine how often the weaker model actually revised its reasoning in a meaningful way.

Third, the two models were not equally strong. Since Gemini 2.5 Flash had a much higher single-agent baseline, the heterogeneous gains may partly reflect capability transfer rather than architectural diversity. A stronger test of the diversity hypothesis would compare two models with similar single-agent accuracy but different training pipelines or architectural properties.

Finally, our evaluation is limited to MATH-500 Level 4 and Level 5 problems. These results may not generalize to factuality tasks, coding, common sense reasoning, or domains where both models have similar weaknesses.

6 Debate Diagnostics and Cost Analysis

Beyond final accuracy, we compute diagnostic metrics from the per-sample debate transcripts: each agent’s Round 1 answer, Round 2 answer, the final

selected answer, and the gold answer. We measure disagreement rate, agreement accuracy, debate contribution, and cost/latency to better understand when debate helps and when it fails.

Disagreement and agreement. Disagreement rate measures how often the two agents end with different final answers, while agreement accuracy measures how often their shared answer is correct when they agree. In the heterogeneous setup, the automatic evaluator initially flagged 19 disagreements. Manual inspection showed that 11 were mathematically equivalent answers written in different formats, such as degree symbols, whitespace differences, or units. After correcting these cases, the heterogeneous setup had 8 genuine disagreements out of 100 samples, giving a true disagreement rate of 8.0%.

The agents genuinely agreed on 92 out of 100 samples. Their accuracy on these agreed samples was 91.4%, while accuracy on genuine disagreements was only 37.5%. This suggests that agreement is a useful confidence signal: when the two models converge to the same answer, the answer is much more reliable than when they disagree.

Debate contribution. We also measure whether debate fixes or breaks Agent 0’s original answer. Debate is counted as helping when Agent 0’s Round 1 answer is wrong but the final selected answer is correct. It is counted as hurting when Agent 0’s Round 1 answer is correct but the final selected answer is wrong.

In the heterogeneous setup, debate helped on 1.0% of samples and hurt on 0.0%. This final-answer effect is small because Agent 0 is Gemini 2.5 Flash, the stronger model, so there are fewer opportunities for the final selected answer to improve. The more important effect appears in the weaker agent’s individual improvement: GPT-4o-mini increased from 54.0% in Round 1 to 84.0% in Round 2 after seeing Gemini’s response. This suggests that heterogeneous debate improves the weaker agent’s post-debate answer quality, even when the final selected answer remains close to the stronger model’s solo baseline.

Cost and latency. Debate is more expensive and slower than single-agent inference because it requires multiple sequential model calls. Table 3 summarizes the average cost and latency per sample.

Table 3: Average API cost and latency per sample.

Setup	Cost	Latency
Single OpenAI	\$0.00049	14.0s
Single Gemini	\$0.01223	18.9s
Hom. OpenAI	\$0.00284	69.2s
Hom. Gemini	\$0.03600	70.5s
Heterogeneous	\$0.02210	68.6s

Heterogeneous debate costs about \$0.02210 per sample and takes 68.6 seconds on average. This is more expensive than either single-agent setup, but cheaper than homogeneous Gemini debate. Its latency is similar to the other debate setups because all debate conditions require multiple sequential model calls.

Table 4: Debate diagnostic metrics across debate setups.

Metric	Hom. OAI	Hom. Gem.	Hetero.
Disagree rate	31.0%	10.0%	8.0%
Agree acc.	71.0%	91.1%	91.4%
Disagree acc.	29.0%	50.0%	37.5%
Any agent correct	62.0%	91.0%	87.0%
Both agents wrong	38.0%	9.0%	13.0%
Helped vs. R1	10.0%	7.0%	1.0%
Hurt vs. R1	2.0%	2.0%	0.0%

Overall, these diagnostics show that final accuracy alone does not fully explain the effect of debate. In the heterogeneous setup, final accuracy remains close to Gemini’s solo baseline, but agreement is a strong confidence signal and GPT-4o-mini improves substantially after seeing Gemini’s response. This suggests that heterogeneous debate may be most useful for improving weaker agents or identifying high-confidence answers through agreement.

7 Conclusion

This project tested whether heterogeneous model debate improves over homogeneous debate in multi-agent LLM deliberation. On difficult MATH-500 problems, homogeneous debate produced only small gains for both GPT-4o-mini and Gemini 2.5 Flash. In contrast, heterogeneous debate substantially improved GPT-4o-mini’s post-debate accuracy, bringing it close to Gemini’s performance.

These results suggest that cross-model deliberation can help when different models contribute different reasoning paths. However, they do not fully isolate architectural diversity from model capability differences. Because Gemini was much stronger than GPT-4o-mini on this benchmark, some of the improvement may reflect the weaker model being

exposed to a stronger model’s answer. Future work should test three-agent setups for majority voting, compare models with similar baseline accuracy, analyze transcripts to separate copying from reasoning, and evaluate additional benchmarks where both models have room to improve.

8 Contribution Statement

Dilan Doshi contributed to experiment design, implementation, evaluation, and the creation of deliverables. Jacob Snodgrass contributed to project conceptualization, experimental design, and report integrity. Arian Dehnavizadeh contributed to analyzing the results of the experiment, putting together the data tables, and creating metrics. Dhruv Patel contributed to running and evaluating multi-agent debate experiments. Shubhan Yenuganti contributed to building the multi-agent debate architecture and evaluation framework.

All team members equally contributed to the final presentation and report, as well as the mid-quarter report and presentation.

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